Radian Measure and Arc Length

Converting between degrees and radians

1 Convert each angle to radians, in exact form:

a $45°$ **b** $120°$ **c** $270°$ **d** $330°$

2 Convert each angle to degrees. Round to one decimal place where needed:

a $\frac{\pi}{6}$ **b** $\frac{5\pi}{4}$ **c** $\frac{7\pi}{3}$ **d** 3.5 rad

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Arc length and sector area

3 A circle has radius 6 cm. Find the exact length of the arc subtended by a central angle of $\frac{2\pi}{3}$.

4 A sector of a circle of radius 10 cm has central angle $\frac{\pi}{5}$. Find its exact area.

5 A wheel of radius 30 cm rolls along the ground without slipping.

a Through what angle, in radians, does the wheel turn when it travels 1.5 m?

b Express this angle to the nearest tenth of a degree.

6 A pendulum of length 80 cm swings through an angle of $25°$. Find the length of the arc traced by its tip, to the nearest tenth of a centimetre.

Angular and linear speed

7 A carousel horse is 4 m from the center and completes one full rotation every 20 seconds.

a Find the angular speed of the horse, in radians per second.

b Find the linear speed of the horse, in m/s.

8 A car's wheel has radius 35 cm and spins at 400 revolutions per minute. Find the car's speed in km/h, to one decimal place.

9 Two gears mesh together: the smaller gear has radius 5 cm and spins at 300 rpm. If the larger gear has radius 15 cm, find its rotational speed in rpm (the linear speed at the point of contact is the same for both gears).

Coterminal angles and reference angles

10 Find a positive coterminal angle (between 0 and 2π) for each:

a $\frac{9\pi}{4}$ **b** $-\frac{\pi}{3}$ **c** $\frac{13\pi}{6}$

11 Find the reference angle for each angle:

a $\frac{5\pi}{6}$ **b** $\frac{7\pi}{4}$ **c** $\frac{4\pi}{3}$